1 Dataset preparation

Study workflow

- Replication
- Analytical correction
- Preprocessing and fitting

Analytical inputs

- Fitted model
- Aligned matrices and labels
- Features and settings
- Optional retraining callback

2 Standardised XAI framework

Configuration

- Shared configuration interface
- Data, model, feature definitions
- Methods, scoring, seeds, and output paths

`run_pipeline(config)`

- Orchestrates enabled stages
- Maintains the shared experiment state
- Records runtime and provenance

Experiment state and exporter

Global

- Native, SHAP, permutation
- Rank agreement
- Top-k overlap

Local

- Correct, incorrect, borderline cases
- Repeated-seed LIME
- Seed-stability assessment

Stability (optional)

- Bootstrap refitting
- Permutation-rank stability
- Pairwise Spearman ρ

3 Research outputs

Reproducible exports

- Model artefacts
- Tables and metrics
- Runtime and provenance
- Configuration metadata

Cross-dataset synthesis

- Compares rankings and stability
- Assesses runtime across datasets
- Synthesises findings